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General Knowledge
World Geography

STUDY NOTES

GEOMORPHOLOGY: LANDFORMS(GLACIAL)

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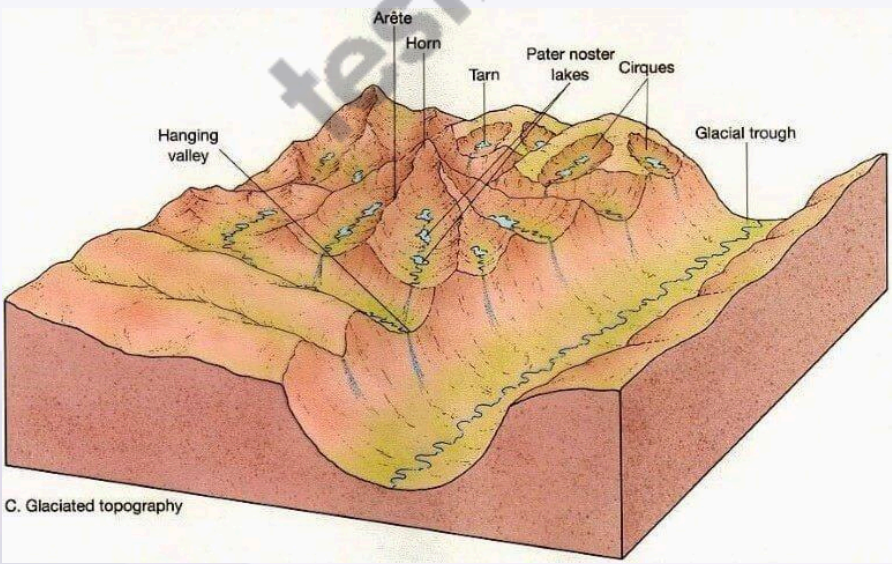
Glacial Landforms

- A glacier is a moving mass of ice at speeds averaging a few meters a day.
- Types of Glaciers: **continental glaciers, ice caps, piedmont glaciers and valley glaciers**.
- The continental glaciers are found in Antarctica and Greenland. The biggest continental ice sheet in **Iceland.**
- Ice caps are the covers of snow and ice on mountains from which the valley or mountain glaciers originate.
- The piedmont glaciers form a continuous ice sheet at the base of mountains as in southern Alaska.
- The valley glaciers, also known as Alpine glaciers, are found in higher regions of the Himalayas in our country and all such high mountain ranges of the world.
- The largest of Indian glaciers occur in the Karakoram range, viz. **Siachen (72 km),** while Gangotri in Uttar Pradesh (Himalayas) is 25.5 km long.
- A glacier is charged with **rock debris** which are used for erosion activity by moving ice.
- A glacier during its lifetime creates various landforms which may be classified into erosional and depositional landforms.

Glacial Erosional Landforms



C. Glaciated topography

Cirque/Corrie

- Hollow basin cut into a mountain ridge.
- It has a steep sided slope on three sides, an open end on one side and a flat bottom.
- When the ice melts, the cirque may develop into a **tarn lake.**

Glacial Trough

- Original stream-cut valley, further modified by glacial action.
- It is a 'U' Shaped Valley. It is at a mature stage of valley formation.
- Since glacial mass is heavy and slow-moving, erosional activity is uniform – horizontally as well as vertically.
- A steep-sided and flat bottomed valley results, which has a 'U' shaped profile.

Hanging Valley

- Formed when smaller tributaries are unable to cut as deeply as bigger ones and remain 'hanging' at higher levels than the main valley as **discordant tributaries**.
- A valley carved out by a small tributary glacier that joins with a valley carved out by a much larger glacier.

Arete

- Steep-sided, sharp-tipped summit with the glacial activity cutting into it from **two** sides.

Horn

- The ridge that acquires a 'horn' shape when the glacial activity cuts it from **more than two sides.**

D-Fjord

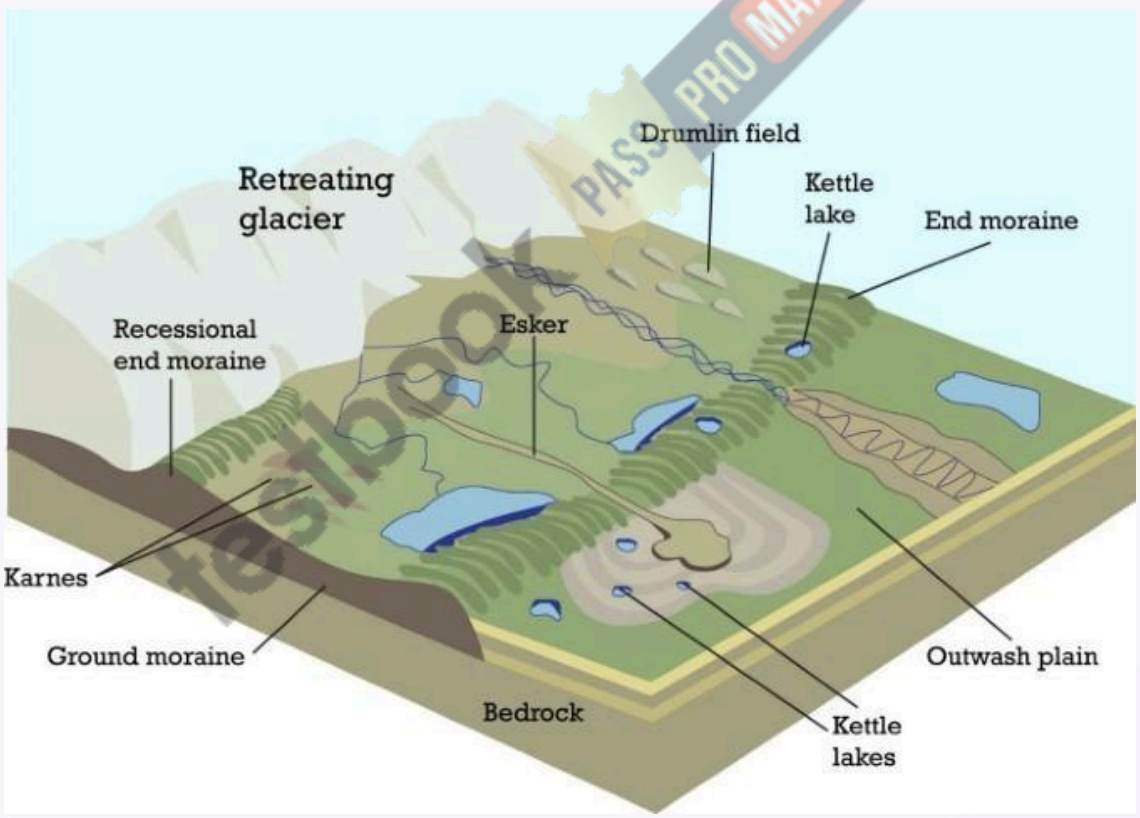
- Steep-sided narrow entrance-like feature at the coast where the stream meets the coast.
- Fjords are common in **Norway, Greenland and New Zealand**.

Note - Why are the world's highest mountains at the equator?

- Ice and glacier coverage at lower altitudes in cold climates is more important than the collision of tectonic plates. (Glacial erosion is very strong because of the huge boulders of rocks carried by the glacial ice that graze the surface. Through ice moves only a few meters a day, it can take along huge rocks that can peel the outer layers.)
- Scientists have solved the mystery of why the world's highest mountains sit near the equator.

- Colder climates are better at eroding peaks. In colder climates, the snowline on mountains starts lower down, and erosion takes place at lower altitudes.
- In general, mountains only rise to around 1,500m above their snow lines, so it is the altitude of these lines — which depends on climate and latitude — which ultimately decides their height.
- At low latitudes, the atmosphere is warm, and the snowline is high. Around the equator, the snowline is about 5,500m at its highest, so mountains get up to 7,000m.
- There are a few exceptions (that are higher), such as Everest, but extremely few.
- When you then go to Canada or Chile, the snowline altitude is around 1,000 m, so the mountains are around 2.5km.

Glacial Depositional Landforms



Outwash Plain

- When the glacier reaches its lowest point and melts, it leaves behind a stratified deposition material, consisting of rock debris, clay, sand, gravel etc. This layered surface is called till plain or an outwash plain.

Esker

- Winding ridge of un-assorted depositions of rock, gravel, clay etc. running along a glacier in a till plain.
- The eskers resemble the features of an embankment and are often used for making roads.

Kame Terraces

- Kame terraces form when sediment accumulates in ponds and lakes trapped between lobes of glacier ice or between a glacier and the valley side.

Drumlin

- Inverted boat-shaped deposition in a till plain caused by deposition.

Kettle Holes

- Formed when the deposited material in a till plain gets depressed locally and forms a basin.

Moraine

- The general term applied to rock fragments, gravel, sand, etc. carried by a glacier.
- Depending on its position, the moraine can be ground moraine and end moraine.